

L05 - Analyzing "Arrival" Through the Lens of NLP

Team Reflections Journal

Mary Ann:

Watching *Arrival* helped me understand how the progression of NLP can be compared to the way Louise learns the Heptapods' language. She begins with simple words, objects, and explicit associations, which reminded me of rule-based NLP. As she collects more examples and begins recognizing patterns, the process becomes more similar to statistical and deep-learning approaches. I found the Heptapods' circular written language especially interesting because the meaning seems to be represented as a complete structure instead of as individual words arranged in a sequence.

The "weapon" versus "tool" interpretation also showed me how important context is in NLP. A translation can technically identify a possible meaning but still fail to capture the intended meaning. The different countries reacted differently because they interpreted the same information through different assumptions and cultural perspectives. The movie reinforced for me that language understanding requires more than translating individual words. It requires context, patterns, and an understanding of how meaning is organized. The connection between the Heptapod language and Louise's changing perception of time also gave me a better understanding of the concept of linguistic relativity and the idea that language can influence how people organize and understand information.

Cesar:

One of the main things I learned from *Arrival* is how difficult it would be to learn a completely unfamiliar language when there is no dictionary, previous translation, or set of examples to work from. Louise cannot begin by translating complex sentences. She has to start with simple words, names, objects, and visual information and gradually establish connections between the human language and the Heptapods' symbols.

This helped me see the importance of context and observation in NLP. Understanding a language is not only about identifying individual words or symbols. The situation in which they are used and the overall meaning of the communication are also important. The movie showed how researchers could collect examples, look for patterns, test possible meanings, and gradually improve their understanding. My main takeaway is that technology can assist with language analysis, but meaningful communication still depends on understanding context and observing how language is actually being used.

Luiz:

Arrival made me think about several NLP challenges that I have also experienced personally while learning and using another language. One of the biggest is ambiguity. A word or symbol can have more than one meaning, and without enough context it can be difficult to know which meaning is

correct. Louise experiences this problem throughout the movie, especially when the Heptapods' message is interpreted using the word "weapon." Different groups react differently because translating a word does not automatically mean that the intended message has been understood.

The movie also reinforced the importance of understanding an entire sentence or situation instead of translating one word at a time. This connects with my own experience learning English, where understanding the context of a sentence often worked better than trying to translate every individual word. I have also seen similar problems when NLP models work across languages. In a previous project, a model had difficulty correctly identifying emotion in Portuguese because it misunderstood important words and their context. These experiences helped me connect the movie to real NLP problems involving ambiguity, culture, language structure, and interpretation. *Arrival* showed me that computers must eventually understand not only the words people use, but what people actually mean.

Nichelle:

While watching *Arrival*, I became most interested in how Louise gradually creates shared meaning rather than simply "translating" an alien language. She begins with written words such as **HUMAN**, physically identifies herself, uses gestures, associates names with individuals, and later connects statements such as "IAN WALKS" with observable actions. I saw this as semantic grounding: both sides first have to establish what a symbol refers to before they can attempt more complex communication. The use of writing, gestures, movement, visual symbols, and behavior also made the communication process clearly multimodal.

I was especially interested in the scenes where Louise analyzes the structural differences among the circular logograms and later uses a computer system to assemble symbols in the Heptapods' language. At the time, I connected these scenes to feature extraction, pattern recognition, representation learning, and the use of technology to identify structure in unfamiliar language. I also found it interesting that the Heptapods' written language did not correspond directly to speech and that their logograms appeared to represent meaning as complete units rather than as a sequence of sounds.

The "offer weapon" scene was the clearest example to me of why context and human judgment remain essential. The problem was not simply translating a symbol; it was deciding what the symbol actually meant in context. Louise challenged the interpretation of "weapon" and considered whether the intended meaning might instead be closer to "tool." This showed how a technically plausible translation can still be misleading if the broader intent, culture, and communication context are not understood.

One thought that developed strongly for me while watching was how different the story might be if *Arrival* were made today. The movie was released in 2016, just before the Transformer architecture was introduced in 2017 and before the rapid growth of modern multimodal AI. I began thinking about how current systems could analyze speech patterns, visual symbols, repeated structures, contextual relationships, and shared data from multiple landing sites much faster than the researchers in the movie. At the same time, I kept coming back to the idea that faster pattern recognition would not automatically solve the problem of meaning. An AI system could still produce a confident but incorrect interpretation when there is no bilingual corpus, dictionary, or verified ground truth.

After Professor Rao's submission comments, I saw that this line of thinking closely overlaps with the five technology areas he identified: speech, computer vision, token-based language pipelines, RNNs/Transformers, and human-in-the-loop evaluation. Those categories were not part of my original wording while watching the film, but my analysis was already moving toward the broader question of how present-day AI could support—and potentially complicate—the communication process.

My biggest takeaway from *Arrival* is that advances in NLP and AI can greatly improve how quickly we detect patterns and generate possible interpretations, but *pattern recognition is not the same as understanding meaning*. Context, uncertainty, and human evaluation remain critical when the meaning itself is still being discovered.